

**Experimental Methods in Computer Science  
(and Informatics Engineering)**  
2025-2026

**Training exercises - second written test, December 13<sup>th</sup>, 2024**

This set of training exercises is meant to help students in the preparation for the second written test scheduled to 13 May 2026. If, for any reason, you conclude that it is not possible to answer a given question explain what is wrong and indicate the reason why the question cannot be answered.

- 1) The following summary table shows the execution time results of a benchmark workload using four different GPUs.
  - a) Do you think there is any difference in the performance of the GPUs? Consider a 90% confidence level in your response.
  - b) If the goal were to determine the fastest GPU, what would you do?

| Compiler | Number of runs | Exec. time mean | Exec. time standard deviation |
|----------|----------------|-----------------|-------------------------------|
| GPU-1    | 7              | 13.25           | 1.243                         |
| GPU-2    | 10             | 12.69           | 2.033                         |
| GPU-3    | 11             | 13.61           | 1.795                         |
| GPU-4    | 8              | 16.43           | 2.318                         |

- 2) (Solved in the PL classes) The concessionaire of the A1 highway installed a set of surveillance cameras at strategic points on the A1 with the aim of detecting drivers with abnormal behaviors (excessive speed, too slow, driving in the wrong direction, sudden changes in trajectory, etc.). The images recorded by the cameras (which are remotely stored on a server) are analyzed in real-time by an artificial intelligence system that detects various types of behaviors by drivers that are considered abnormal or dangerous. Unfortunately, as is often the case in such systems, the detection of the desired behavior patterns is far from perfect, so the system may exhibit a relatively high number of false positives, incorrectly identifying some drivers' correct behavior as abnormal.

The concessionaire decides to compare two artificial intelligence models from two companies, called SmtEye and XVision, in order to select the one with less false positives. These two models have similar functionality and have been previously trained to identify drivers with abnormal behaviors with the same set of videos. To perform the experiment, the concessionaire selected a team of three operators that analyzed the video recordings of all cases of abnormal behavior detected by any of the systems for two weeks, from October 28, 2024, to November 11, 2024. Not surprisingly, the three operators identified a reasonable number of false positives in both systems. It is important to note that the distinction between abnormal and normal behavior is not always easy, but the fact that the same images are evaluated by three operators always allows for an unequivocal decision by majority. The table below shows the number of false positives found on each day of the evaluated period for both systems:

|               | Mon | Tues | Wed | Thur | Sex | Sat | Su | Mon | Tues | Wed | Thur | Sex | Sat | Su |
|---------------|-----|------|-----|------|-----|-----|----|-----|------|-----|------|-----|-----|----|
| <b>SmtEye</b> | 7   | 9    | 12  | 6    | 15  | 12  | 5  | 6   | 13   | 8   | 12   | 7   | 8   | 8  |
| <b>XVisio</b> | 6   | 6    | 7   | 7    | 8   | 5   | 4  | 3   | 7    | 5   | 7    | 8   | 4   | 3  |

Which system is the best concerning the number of false positives. Consider a confidence level of 90%. Furthermore, considering the nature of the data, it is risky to assume that the two samples of data follow a normal distribution.

- 3) (Similar to the previous exercise but with some relevant differences) The concessionaire of the A1 highway installed a set of surveillance cameras at strategic points on the A1 with the aim of detecting drivers with abnormal behaviors (excessive speed, too slow, driving in the wrong direction, sudden changes in trajectory, etc.). The images recorded by the cameras (which are remotely stored on a server) are analyzed in real-time by an artificial intelligence system that detects various types of behaviors by drivers that are considered abnormal or dangerous. Unfortunately, as is often the case in such systems, the detection of the desired behavior patterns is far from perfect, so the system may exhibit a relatively high number of false positives, incorrectly identifying some drivers' correct behavior as abnormal. The concessionaire decides to evaluate the installed system, called XEye V1, by having a team of three operators analyze video recordings of all cases of abnormal behavior detected by the system for two weeks, from October 28, 2024, to November 11, 2024. Not surprisingly, the three operators identified a reasonable number of false positives. It is important to note that the distinction between abnormal and normal behavior is not always easy, but the fact that the same images are evaluated by three operators always allows for an unequivocal decision by majority. The table below shows the number of false positives found on each day of the evaluated period:

| XEye V1 | Mon. | Tues. | Wed. | Thur. | Sex. | Sat. | Sun | Mon. | Tues. | Wed. | Thur. | Sex. | Sat. | Sun |
|---------|------|-------|------|-------|------|------|-----|------|-------|------|-------|------|------|-----|
|         | 7    | 9     | 12   | 6     | 15   | 12   | 5   | 6    | 13    | 8    | 12    | 7    | 8    | 8   |

As the concessionaire considers these numbers of false positives to be too high, it decided to request the company that built the system to improve the detection algorithm and introduce some changes in image processing. The result is a new version of the system called XEye V2. This system was tested using the same images recorded for the two weeks period, to ensure a good comparison between the two versions. The table below shows the results of version V2:

| XEye V2 | Mon. | Tues. | Wed. | Thur. | Sex. | Sat. | Sun | Mon. | Tues. | Wed. | Thur. | Sex. | Sat. | Sun |
|---------|------|-------|------|-------|------|------|-----|------|-------|------|-------|------|------|-----|
|         | 6    | 9     | 6    | 7     | 8    | 9    | 6   | 4    | 8     | 7    | 11    | 8    | 5    | 7   |

Considering that the number of cars is similar over the different days analyzed, and that the distribution of abnormal behaviors approximately follows a normal distribution, can it be confidently stated with 95% confidence level that version V2 is better than version V1 in terms of false positives?

- 4) A Bank has developed a new database system with the goal of improving the speed of typical web banking transactions done by the Bank customers. There are several types of transactions (e.g., viewing account balances, funds transfers between the customer's accounts, viewing transactions in a given period indicated by the customer, etc.) The response time of each transaction depends on several factors related to both the queries issued by the customer and the load of the system. In order to assess the new database response, the Bank asked its Information Systems Department to perform an experiment to test the speed of the new database and compare the speed with the current database. The experiment includes a set of representative accounts with typical transactions profiles to assure that the measurements of the speed in both databases (the new one and the already existing) are meaningful.
- a) State the different elements of the experiment: i) problem definition, ii) variables, liii) levels, and iv) hypothesis. Justify your choices.

- b) In very few cases it was observed the response time of the transactions was several times larger than the average response time. Explain the choices you have to deal with this problem.
- 5) Complete the following two-way ANOVA table for two factors (F1 e F2). In column “Decision”, you should write your decision with respect to the null hypothesis. Justify your decisions and present all the required calculations. Assume a significance level of 5%.

|                    | SS       | df | MS | F value | Decision |
|--------------------|----------|----|----|---------|----------|
| <b>F1</b>          | 835.75   | 2  |    |         |          |
| <b>F2</b>          | 1426.04  | 1  |    |         |          |
| <b>Interaction</b> | 14.58    |    |    |         |          |
| <b>Within</b>      | 1270.25  |    |    |         |          |
| <b>Total</b>       | 3546.625 | 39 |    |         |          |

- 6) In an experiment, you suspected that two variables are correlated. You applied a Pearson’s test to samples of the two variables and the result for the Pearson’s coefficient was  $r = 0.989$ . Based on this result you concluded that there is a strong positive correlation between the two variables. But you are interested in knowing if this correlation does in fact imply causation.
- a) Why is it important to know that correlation also implies causation? In another words, what can you do if you are sure that the high positive correlation also implies causation for the above-mentioned variables?
- b) Explain the type of analysis you would perform to confirm whether the correlation also implies causation or not.
- 7) You are trying to improve the average execution time of a long aggregation query in a decision support database (normally know as data warehouses). These types of queries typically scan big portions of the main table of the data warehouse (called the fact table) and filter the registers using several satellite dimension tables. As the execution time of such queries can take minutes or even hours, all the improvements are very welcome. After generating a new set of indexes and materialized views (the typical “tricks” used to speed up such queries) you tried several variants of the query, and you observed some reduction in the average execution time. However, when comparing the samples (i.e., execution time measurements) obtained before and after the introduction of the new indexes and materialized views, you concluded that you could not reject the null hypotheses that the average execution time is the same in both scenarios with 90% of confidence. In any case, based on your calculations, you would be able to reject the null hypothesis with 70% of confidence. In these circumstances, answer to the following questions:
- a) Do you think you should leave the data warehouse as it was, or should you consider the new set of indexes and materialized views as the next production version of the data warehouse? Justify your answer and explain the situation in terms of type of errors in hypothesis testing.
- b) If instead of improving the execution time of a database query for decision support purposes, the database query wes used in a system to support the schedule of

ambulances and rescue team in the medical emergency response to calls to the 112 national number, do you think that would change your answer?

- 8) The general intuition is that female code inspection teams are better than male inspection teams, when performing classic Michel Fagan's code inspections (see, for example, [https://en.wikipedia.org/wiki/Fagan\\_inspection](https://en.wikipedia.org/wiki/Fagan_inspection) to know the basics about Fagan's software inspections). In other words, the general intuition is that female inspection teams can find more bugs in code inspections than male teams. However, as simple intuition is not enough, your company has decided to find out whether there is a difference in the performance of male and female code inspection teams.

Suppose you have been in charge of designing such experiment and you recorded the number of bugs found by male and female inspection teams in the inspections performed in your company in the last month. The code units inspected represent a variety of the code developed by the company and all the code units have been developed using the same software development methodology (i.e., the chances of having residual bugs are identical).

The following table shows the number of bugs found in the code inspection performed in the last month:

|                     |   |   |   |   |   |   |   |   |
|---------------------|---|---|---|---|---|---|---|---|
| <b>Male teams</b>   | 3 | 5 | 2 | 3 | 2 | 3 | 8 | 5 |
| <b>Female teams</b> | 6 | 4 | 9 | 6 | 7 | 8 | 7 |   |

Based on these results, can you state, with 95% of confidence, that the performance of male and female code inspection teams is different? Note that it is not realistic to assume that the number of bugs found in code inspections follows a normal (or close to normal) distribution.

- 9) The NVIDIA® DGX-1™ is a GPU-based hardware accelerator for deep learning processing (in fact, it's a compact supercomputer). Your company has a contract with a major European automaker and is tasked with developing and testing the collision avoidance system. This system uses cameras, ultrasonic sensors, and front radar to detect objects in the car's surroundings. All of this is used in the overall autopilot system that will equip the car, and it uses the NVIDIA® DGX-1™ to process data in real-time. However, your responsibility in the project is centered only on the collision avoidance system. Since this system is seen as an auxiliary system in level 3 autonomous driving mode, the responsibility still lies with the driver, who must be prepared to take control of the car whenever necessary. In other words, the collision avoidance system is expected to be able to detect a high number of situations, but perfect performance is not yet required, as it would be in level 5 autonomous driving (in a future that we still don't know when will come).

To evaluate the collision avoidance system, it is necessary to design an experiment that uses previously recorded driving scenarios (containing recorded images and sensor signals), allowing the assessment of system performance through the replay of these pre-recorded scenarios. These scenarios are grouped into 5 benchmarks, each representing different obstacle detection situations on the road (e.g., detecting children, detecting small animals, detecting tree branches, etc.). Each of the 5 benchmarks has 20 scenarios, but only 10 of these scenarios involve situations that should trigger the activation of the collision avoidance system. From the perspective of the experiment, it is considered that the collision avoidance system has performed as expected if, for each scenario, it manages to detect the obstacle in time to avoid a collision or if it does not incorrectly detect obstacles in scenarios where there are none on the road.

The experiment also aims to evaluate various operating modes of the collision avoidance system, which in practice correspond to different Deep Neural Network (DNN) algorithms for obstacle detection.

In these circumstances, answer the following questions:

- a) What type of hypothesis test would you use in the described scenario? Justify your answer.
  - b) Dependent variables.
  - c) Independent variables.
  - d) Examples of possible levels for the independent variables.
  - e) Provide three examples of hypotheses that could be tested. State the hypotheses in full.
- 10) Software inspections proposed by Michael Fagan is a technique to find errors during software development process. Inspections can be applied to any phase of the software development process, but Fagan's inspections are mainly used to **verify the requirement specifications** (because unlike software code that can be run and tested, requirements cannot be tested automatically). Very briefly, in a requirement inspection a group of inspectors (software developers) examine the document describing the requirements in order to find possible errors, inconsistencies or missing requirements. It is a collective effort and during an inspection session one of the inspectors (the reader) is actually reading aloud the requirement specification while the others (typically 2 or 3 inspectors) verify the correction and consistency of the requirement description that are being read by the reader. Whenever one of the inspectors thinks there is an error he/she raises the issue and, if the group of inspectors agree that the raised issue is really an error, such error is recorded by one of the inspectors (the recorder). The errors are simply recorded for posterior analysis of the possible solution (i.e., the inspectors do not discuss how the error found can be solved; that will be done afterwards by the authors of the requirement specification).

Your company uses requirement inspection very often. But recently the company has decided to introduce several changes in the Fagan's inspection process and create a new inspection methodology called FaganT++. In order to verify that the new methodology is superior to the traditional Fagan's inspections, the company selected a group of 10 developers to perform a controlled experiment to compare the two inspection methodologies. The main point under comparison (dependent variable) is the capacity of error detection of each inspection method. Although the inspections also detect some false positives (i.e., record errors that are not real errors) that is not very important, as the authors of the requirements can easily identify them as false positives later on, when they process the results of the inspection to solve the problems found.

The group of 10 developers was asked to inspect two sets of requirement specifications where someone has previously inserted 50 errors (i.e., small modifications in the requirement descriptions that simulate real requirement errors) in each set. The first set was inspected using the Fagan's method and the second set was inspected using the FaganT++. The group of 10 inspectors performed first the inspection using the Fagan's method. After that, the inspectors received specific training during several days and when the inspectors are already proficient in the FaganT++'s method they perform the inspection of the second set of requirements. As mentioned, both sets have been injected with 50 errors.

The following table shows the errors detected by each inspector in the two inspections. In real inspections what is recorded is the performance of the group of inspectors. But in this controlled experiment it was decided to analyze the individual performance of each

inspector. The inspector I8 does not participate in the second inspection (the one using FaganT++) because of health reasons.

| Inspectors →               | I1 | I2 | I3 | I4 | I5 | I6 | I7 | I8 | I9 | I10 |
|----------------------------|----|----|----|----|----|----|----|----|----|-----|
| <b>Fagan inspection</b>    | 33 | 45 | 28 | 32 | 26 | 36 | 40 | 31 | 32 | 42  |
| <b>FaganT++ inspection</b> | 40 | 42 | 34 | 32 | 34 | 40 | 35 |    | 38 | 39  |

Considering these results, can you say that the new the FaganT++ method is better than the classic Fagan's inspections with 95% of confidence? Show the steps you used and your calculations.

- 11) A company wants to choose the fastest cryptographic hash function available and, to do so, will analyze 8 implementations of various types of hash functions. To evaluate all functions on an equal footing, it will use the same set of test vectors (i.e., data sets available on GitHub) with all the functions. Additionally, all tests are conducted on the same machine (i.e., the same hardware and the same operating system). However, since it is important to assess the performance of the functions under different available memory scenarios, each function is tested on the same machine but in various available memory situations. Furthermore, it is also important to study the impact of the size of the generated hashes, as some functions are faster when generating small hashes but become slower when asked to generate longer hashes.

In these circumstances, explain how you would organize an experiment to choose the fastest hash function and present all the relevant elements of the experiment and all the steps for hypothesis testing. It is important to consider the choice of hypothesis testing techniques that take into account the effort (i.e., cost) of the experiment. Naturally, since there is no data (and thus calculations cannot be made), you are expected to provide the elements of the experiment, the techniques to use, and the steps to follow. Justify your response appropriately.

- 12) An informatics engineer is designing an experiment and realizes that he/she has a very high number of independent variables. After some preliminary experiments, they find through a Pearson test that the correlation coefficient between two variables is -0.9986. However, they ultimately decide to keep both independent variables and consider them in the experiments to be conducted. In your opinion, what is the reason (or reasons) for the engineer not discarding one of the variables, given the Pearson value obtained? Explain the situation.

- 13) Assume that you are the system administrator of a large organization and, in this role, you are responsible for the organization's data center and network infrastructure. In recent months, you have noticed strange behavior in one of the storage servers, consisting of an abrupt drop in the server's performance, followed by a return to normal performance after some time (tens of minutes).

The company that supplied the storage server has already analyzed the system (hardware and software) and declared that the storage server is in perfect condition, claiming that the strange behavior may be caused by an interaction with other systems that use the storage server's services.

Since the problem persists, you decided to increase the amount of data recorded in the storage server logs and other related machines (to gather more information that could help identify the problem). After some analysis, you believe you have found an explanation for

the phenomenon. Formally describe how you would confirm that your explanation is correct.

- 14) A company intends to purchase a large-scale server for high-performance computing. Given the various offers from suppliers, the company decided to request a performance test of the proposed server configuration from each supplier. If there are no performance differences between the servers, the purchasing decision will be based on other criteria (i.e., not performance), particularly factors such as cost and maintenance contract terms.

Since execution time depends on the type of program, the company defined a workload consisting of several programs representative of the type of processing the company needs. This workload was made available to each supplier for testing, with instructions that it should be executed multiple times so that the execution time could be calculated as the average of the execution times from each run. The results received from each supplier are shown in the following table:

|                               | Serv. F1 | Serv. F2 | Serv. F3 | Serv. F4 |
|-------------------------------|----------|----------|----------|----------|
| Number of executions          | 7        | 6        | 6        | 7        |
| Average execution time (sec.) | 20,2     | 15,8     | 21,9     | 24,0     |
| Standard deviation            | 5,13     | 5,59     | 4,78     | 3,66     |

- a) Given these results, the company wants to know if there are significant performance differences among the servers, considering a 95% confidence level. Show the steps and calculations performed in your solution.
- b) Regardless of the conclusion found in a), suppose you want to determine whether server F2, which appears to be the fastest, is indeed the fastest server. Explain the approach you would use to find out if F2 is the fastest server (just explain, no need to provide calculations).
- 15) If you have two samples with  $n < 30$  and the data distribution is not normal, what are the consequences of using a two-sample T-test instead of a non-parametric method? Answer as thoroughly as possible. To make your answer clearer, it is suggested that you list the consequences one by one, rather than using a single sentence.
- 16) A group of individuals representing typical users participates in the evaluation of an improved version of a sports brand's shopping App for iOS and Android phones. The evaluation consists of a set of five tasks (searching and purchasing various items) to be performed in the App, and the result is the total time each user spends completing the tasks. The new interface version (V2) is simpler than the previous version (V1), so it is expected that the time required to perform tasks with interface V2 will be shorter than with V1.

The experiment was organized as follows: the nine participants first used the original App (V1), performed the pre-prepared tasks, and their execution times were measured. The total time taken to complete the five tasks was calculated and recorded for each participant. Later, the same nine participants performed the same tasks using V2. Due to a technical issue, the task execution times of participant P7 could not be recorded. The following table shows the results:

|    | P1  | P2  | P3  | P4  | P5  | P6  | P7  | P8  | P9  | Average | Std. Deviation |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---------|----------------|
| V1 | 6.7 | 4.2 | 6.3 | 9.9 | 5.3 | 7.3 | 5.7 | 9.5 | 8.5 | 7.044   | 1.936          |
| V2 | 4.8 | 7.4 | 5.1 | 4.7 | 6.9 | 5.1 |     | 8.4 | 6.9 | 6.163   | 1.408          |

- a) Based on these results, should V1 be replaced by V2? Consider a 95% confidence level.
- b) Suppose the experiment had the following differences:
  - Participants have no prior knowledge or experience with either App version (V1 or V2).
  - During the tasks, it was never indicated whether the participant was using V1 or V2.
  - For each participant, it was randomly decided whether they performed tasks first in V1 and then in V2 or vice versa.
- c) Would this new experimental setup be better than the one previously described? Justify your answer. Also, suggest what changes you would make to evaluate whether V1 should be replaced by V2 (only explain what you think should be changed; there is no need to repeat calculations).

- 17) Your company intends to choose a lossless data compression algorithm and is considering four well-known algorithms: Gzip (Deflate), Bzip2 (Burrows–Wheeler), LZMA (Lempel–Ziv–Markov chain), and Zstandard.

Since compression rates depend on the nature of the data, the company decided to conduct an experiment with a significant sample of files from the company's data repository to determine whether there are significant differences in compression rates for the different algorithms. The dataset includes eight files and the experiment consists of measuring the compression rate of each algorithm for each file, in order to calculate the average compression rate of each algorithm and the standard deviation. The results are the following ones:

| Algorithm | Average compression rate | Std. deviation |
|-----------|--------------------------|----------------|
| Gzip      | 5.76                     | 1.06           |
| Bzip2     | 6.07                     | 0.98           |
| LZMA      | 7.08                     | 1.28           |
| Zstandard | 5.56                     | 1.07           |

The compression rate values are presented in terms of the factor by which the original size must be divided to obtain the size after compression. For example, for Gzip, the average compression factor is 5.76. This means that when compressing a 100MB file, the compressed file will have a size of  $100\text{MB}/5.76 = 17.36\text{MB}$ . Obviously, this is an estimate based on the average compression factor.

The speed of compression and decompression is not very important for the company, as the intended function is primarily for data archiving. Therefore, if the compression rates of the different algorithms are similar, the company will decide to purchase the commercial version of the cheapest algorithm (i.e., the decision is based solely on cost).



Based on a 95% confidence level, determine whether there are differences in the compression rates among the algorithms. Show your calculations, explain the process you used, and present your conclusion.

- 18) Suppose you want to choose a real-time operating system for use in an industrial control application. An important criterion is the stability of the operating system's code, and it is advisable to use an operating system whose code is as stable as possible. One way to evaluate the stability of an operating system's code is by counting the number of patches the operating system has had in each version. Patches introduce small changes to the code, generally with the aim of fixing defects or surgically modifying functionalities. The fewer patches, the more stable the code. It is important to consider that applying patches in an industrial control environment is always challenging, so it is essential that the operating system is stable and not expected to require many patches.

The following table shows the number of patches for two operating systems under consideration across successive available versions. Operating System A is currently at version 7, while Operating System B is at version 9. Given the nature of the data, it is not expected to follow a normal distribution.

|                    | V1 | V2 | V3 | V4 | V5 | V6 | V7 | V8 | V9 |
|--------------------|----|----|----|----|----|----|----|----|----|
| Operating system A | 12 | 8  | 7  | 12 | 5  | 6  | 5  | -  | -  |
| Operating system B | 10 | 6  | 7  | 7  | 9  | 5  | 4  | 3  | 6  |

Considering a significance level of  $\alpha = 10\%$ , determine whether there are differences in stability (i.e., the number of patches across different versions) between the two operating systems, showing all the steps and calculations. If there is no significant difference, then you will know that the choice of the operating system should be based on other criteria (e.g., functionality, price, etc.) rather than greater or lesser stability.

- 19) A computer game production company aims to study in-depth the mechanisms of pleasure/frustration associated with computer games to optimize them in a way that generates the maximum desire to play. The idea is to study brain activity at the moment the player makes a mistake or wins a game, using an fMRI (functional magnetic resonance imaging) device, as neurologists know that certain brain regions are associated with sensations of pleasure and frustration. By measuring the activity in these regions, the company believes it can evaluate the game steps that elicit the most pleasure or frustration.

For this purpose, the player must be inside an fMRI while playing so that brain activity can be analyzed. Knowing that the space inside an fMRI is very limited and quite artificial, the company fears this might alter the game conditions—for example, causing the player to make more mistakes and lose more frequently. Therefore, before proceeding with the study, the company decided to conduct an experiment to verify whether being inside an fMRI significantly alters the game conditions, particularly increasing the number of mistakes made by the player. In other words, the focus is essentially on the frustration associated with mistakes.

The experiment involved asking a group of volunteers to play the game under normal conditions (i.e., not inside the fMRI) and then play it inside the fMRI. These tests were carried out with a group of 13 players who played the game for 20 minutes in each scenario. The players committed the following numbers of mistakes:

Errors under normal conditions:

| J1 | J2 | J3 | J4 | J5 | J6 | J7 | J8 | J9 | J10 | J11 | J12 | J13 |
|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|
| 23 | 34 | 22 | 39 | 19 | 26 | 29 | 32 | 25 | 21  | 33  | 36  | 34  |

The mean of this sample is 28.60, and the standard deviation is 6.18.

Errors inside the fMRI:

| J1 | J2 | J3 | J4 | J5 | J6 | J7 | J8 | J9 | J10 | J11 | J12 | J13 |
|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|
| 34 | -  | -  | 30 | 29 | -  | 41 | -  | 23 | 43  | -   | 39  | -   |

The mean in this case is 34.14, and the standard deviation is 6.72.

Given the inherent difficulties in conducting tests inside an fMRI machine, the company was only able to complete the experiment with the 7 players indicated in the table above.

Under these circumstances, and considering the company's objectives, can you determine whether performing the experiment inside the fMRI machine affects the number of errors made by each player? Use a 95% confidence level, clearly state the conclusion you reached, and explain the reason(s) for choosing the hypothesis testing method employed.